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ex 9

Problem 1.

Graph construction:

① 81 vertices – one per cell
② edge between 2 vert.s if they share a row, column, or 3x3 box
③. 9 colours = digits 1-9

Boundary conditions

every row, column, and box must use each colour exactly once → no two adjacent vertices share a colour.
pre-filled cells = pre-colour vertices that constrain the remaining colouring.

Problem 2.

1.2. Interations.

initial: $d(S)=0$, all others $=\infty$

interation 1. relax all edges

Vertex	S	E	A	D	C	B
dist	0	8	10	∞	∞	∞

$S \to E: 0+8=8$, $S \to A: 0+10=10$

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interation 2

vertex	S	E	A	D	C	B
dist	0	8	10	9	12	11

E→D: 8+1=9 , A→C: 10+2=12 , A→B: 10+1=11

interation 3

vertex	S	E	A	D	C	B
dist	0	8	5	9	8	9

D→A : 9-4=5 , D→C : 9-1=8 , C→B : 8+1=9

Interation 4.

vertex	S	E	A	D	C	B
dist	0	8	5	9	7	8

A→C : 5+2=7, A→B : 5+1=6, C→B: 7+1=8

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3. final shortest distance from S.

vertex	shortest distance from S
S	0
E	8
A	5
D	9
C	7
B	8

4. why Dijkstra fails here

- Dijkstra algorithm assumes all edge weights are non-negative It greedily picks the vertex with the smallest current distance and never revisits it. with negative edges (e.g. D->A :-4). a vertex already "finalized" by Dijkstra may later be reachable via a shorter path, making

5. Negative cycle

No, after V-1 iterations, running one more produce no further relaxations, distance are stable, Therefore the graph contains no negative cycle